Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store

A Proposal Report for BDM Capstone Project

Submitted by

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Contents

1. Executive Summary

This midterm report presents preliminary findings from a data-driven analysis of *Jhansi Rope Store*, a third-generation family-run utility goods business located in Subhash Ganj, Jhansi, established in 1965. The study addresses inventory management and demand forecasting challenges in a dual-market system serving both B2B (wholesale) and B2C (retail) customers across urban and rural segments. A dataset of 640 authentic sales transactions was systematically collected from the store's manual registers over a 14-week period (July 20–October 31, 2025). It includes customer type, product category, quantity, price, location, and verified profit margins ranging from 4–7%. Data was processed using Python (Pandas, Matplotlib) and Excel for descriptive, trend, and pattern analysis.

Preliminary findings reveal strong seasonal and geographic variations in sales performance. Revenue rises by 127% from the regular period (₹20.56 lakhs) to the festival season (₹45.36 lakhs), confirming the need for demand forecasting. Rural customers contribute 54% of total revenue with 150% higher seasonal sensitivity, highlighting distinct market behavior. ABC analysis identifies Niwar (40%) and Brooms (25%) as fast-moving items generating 65% of revenue. Wholesale sales constitute 65% of transactions with 4–5% margins, while retail achieves 9–12%, reflecting contrasting profitability patterns. Supporting materials include metadata, descriptive statistics, visual dashboards, and an authorization letter validating the study's originality.

2. Proof of Originality

Components of Proof of Originality Submission:

1. Store Photographs (3 images): Capturing the shop, interior, storage area, and billing counter.
2. Transaction Bills (2 images): Original customer bills collected during the data collection period (July–October 2025).
3. Owner Interaction Video: A recorded conversation with the business owner discussing sales patterns, inventory practices, and market trends.The video is in Hindi; please click on the "CC" (Closed Captions) option to view English subtitles.
4. Proof of Originality Document (PDF): A signed declaration confirming that all data were personally collected and not sourced from any external dataset.
5. Dataset of Jhansi Rope Store: The authentic dataset compiled from manual sales registers (July–October 2025), containing 640 verified transaction records used for analysis

Drive Link to Proof of Originality Folder:

https://drive.google.com/drive/folders/1W0_RFasiK4JCnmlbr1bo2_xS_zsepZcU?usp=drive_link

3. Metadata and Descriptive Statistics

3.1 Dataset Structure and Overview

The dataset includes 640 sales transactions from Jhansi Rope Store, recorded over a 14-week period (July 20–October 31, 2025). Each record captures 20 core variables covering customer demographics, product details, sales amounts, and contextual factors such as location and festivals. The data was manually transcribed from authentic store records into Excel and analyzed using Python (Pandas). With no missing values, the dataset demonstrates complete integrity and reliability for both descriptive and predictive analysis.

3.2 Key Variables and Business Relevance

Variable	Definition / Format	Business Relevance
Order_Number	Unique sequential ID (ORD1000–ORD1639)	Enables transaction tracking and reconciliation
Date	Calendar date (YYYY-MM-DD)	**Problem 1:** Seasonal and festival demand forecasting
Customer_ID	Unique customer code (C001–C047)	Enables customer-level trend and repeat purchase analysis
Customer_Name	Registered customer or outlet name	Differentiates major wholesale vs retail customers
Customer_Type	Wholesale / Retail	**Problem 3:** Segmentation for margin and stock planning
Product_ID_SKU	Product code (P001–P010)	Supports stock-level tracking and inventory reconciliation
Product_Category	Niwar, Rope, Broom, Wiper, Sutli	**Problem 2:** Used for identifying fast- and slow-moving items
Product_Type	Specific product description	Provides granularity for product-level insights
Variety	Material variants (Cotton, Plastic, Jute, etc.)	Captures product differentiation and buyer preference trends
Quantity_Sold	Number of units sold per transaction	**Problem 1:** Core metric for demand forecasting
Unit	Unit of measure (Kg, Bundle,	Standardizes stock and sales

	Piece)	data across categories
Unit_Price_Rs	Selling price per unit (₹40–₹142)	Reveals pricing patterns and seasonal variations
Total_Sale_Value_Rs	Quantity × Unit_Price	**Problem 1:** Total revenue, used for seasonal trend analysis
Cost_Price_Per_Unit_Rs	Procurement cost per unit (₹38–₹133)	Validates margin structure for profitability analysis
Total_Cost_Price_Rs	Quantity × Cost_Price_Per_Unit	Enables cost tracking per transaction
Profit_Rs	Sale Value − Cost Value	Direct profitability indicator per order
Profit_Margin_Pct	(Profit / Sale Value) × 100 = 4.03–12.50 %	Confirms 4–7% wholesale and 9–12% retail margin spread
Location	Customer/delivery area (14 within 100 km)	Supports spatial sales analysis (linked to Problem 1 trends)
Area_Type	Urban / Semi-Urban / Rural	Distinguishes customer geography for demand behavior
Festival_Event	Regular, Navratri, Post-Navratri, Diwali Prep/Peak/Final	**Problem 1:** Captures external festival influences on demand

Summary Sheet: Reports total revenue, cost, profit, and margin for the analysis period. Serves as a concise dashboard for overall business performance and validates key findings presented in the report.

Product Analysis Sheet: Aggregates sales, cost, profit, and margin by product category (Niwar, Rope, Broom, Sutli, Wiper), including ABC segmentation to support inventory prioritization and product-level recommendations.

B2B vs B2C Sheet: Compares wholesale (B2B) and retail (B2C) transactions across revenue, cost, profit, margin, and transaction count. Supports business model segmentation, profitability comparison, and strategic planning.

Geographic Analysis Sheet: Summarizes financial and operational metrics by area type—Urban, Semi-Urban, and Rural—to guide inventory allocation, highlight regional and seasonal sales trends, and inform event-based planning

4.3 Numerical Data Analysis

```
df.describe()
```

	Quantity_Sold	Unit_Price_Rs	Total_Sale_Value_Rs	Cost_Price_Per_Unit_Rs	Total_Cost_Price_Rs	Profit_Rs	Profit_Margin_Pct
count	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000	640.000000
mean	82.392188	90.154687	7445.396875	84.394531	7060.940625	384.456250	6.431844
std	46.419610	32.233297	5239.650115	30.368296	5039.715004	209.093809	2.562447
min	9.000000	40.000000	378.000000	38.000000	342.000000	36.000000	4.030000
25%	33.000000	62.000000	2946.750000	59.000000	2646.000000	224.000000	4.550000
50%	88.000000	100.000000	6300.000000	95.000000	5985.000000	330.000000	4.850000
75%	122.000000	113.000000	11210.500000	108.000000	10686.500000	521.625000	9.240000
max	160.000000	146.000000	22101.000000	133.000000	21147.000000	954.000000	12.500000

The dataset of 640 transactions shows that the mean quantity sold per order is **82.4 units** (SD = 46.4), ranging from **9 to 160 units**, indicating high variability. The average unit price is **₹90.15** (₹40–₹146), while total sale value per transaction ranges from **₹378 to ₹22,101**, averaging **₹7,445.40**. The average cost price per unit is **₹84.39**, with a **median profit of ₹330** and an **average margin of 6.4%**, typically between **4.5% and 9.2%**.

These statistics highlight variation in order sizes between **B2B** and **B2C** transactions while confirming stable pricing and sustainable profitability. The analysis supports decisions on **inventory planning, pricing, and demand forecasting**, and identifies **Niwar** as the most sold product and **Wholesale** as the leading transaction type, aligning with observed seasonal trends.

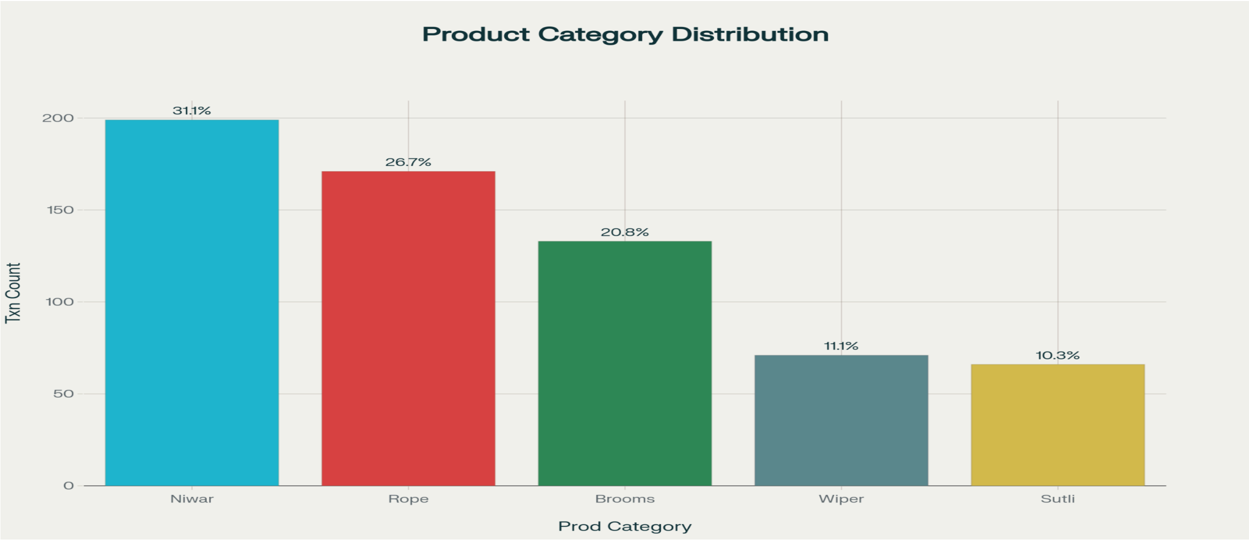
4.4 Categorical Data Analysis

The categorical distribution of transactions offers insights into product mix, customer segmentation, and geographic reach:

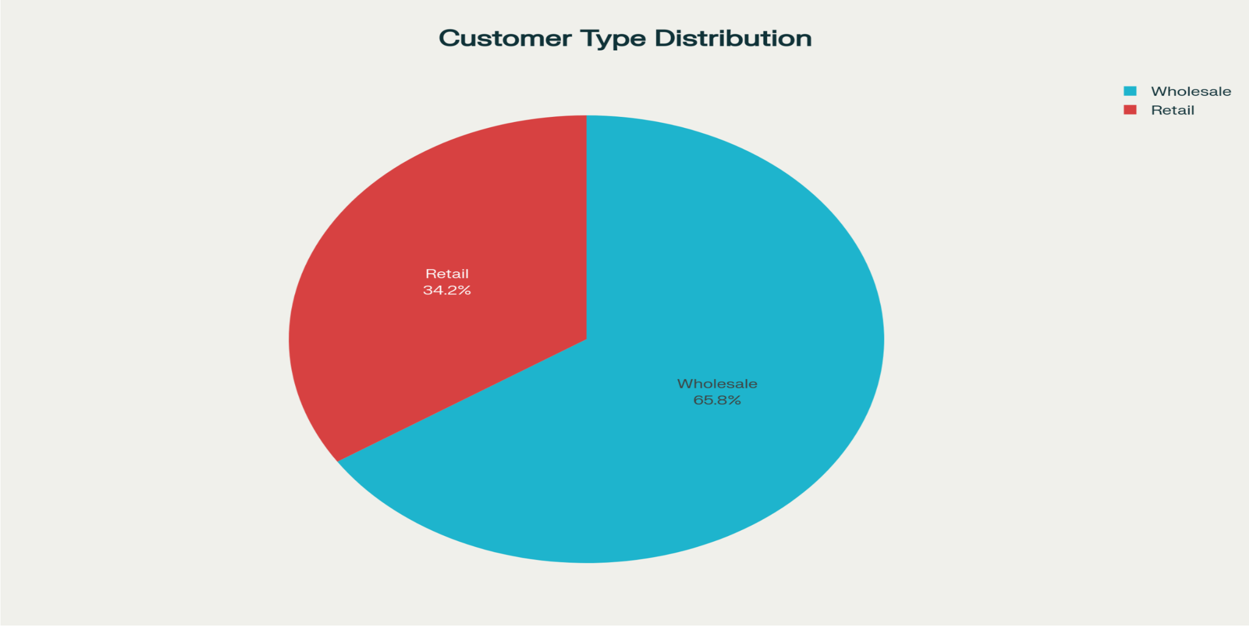
Product Category: Most sales occur in Niwar (31.1%) and Rope (26.7%), identifying them as the store's primary revenue drivers. Secondary categories—Broom, Wiper, and Sutli—contribute to product diversity (see Product Category Bar Chart).

Customer Type: Wholesale (B2B) transactions account for 65.8%, while Retail (B2C) comprises 34.2%, showing a predominantly B2B structure with a growing retail base (see Customer Type Pie Chart).

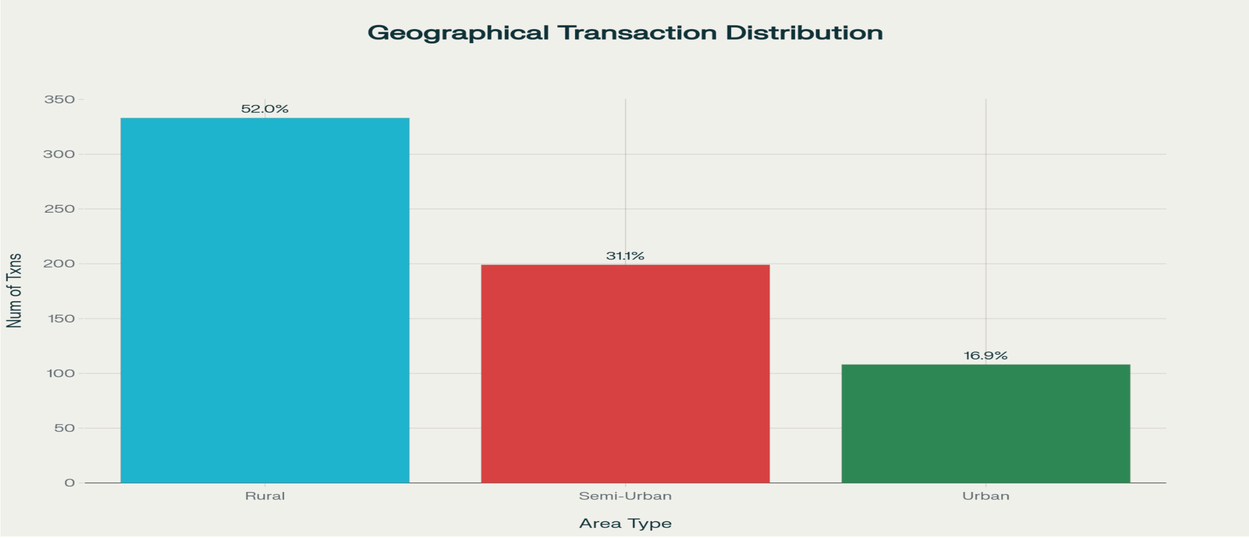
Geographic Distribution: Rural markets dominate with 52% of transactions, followed by Semi-Urban (31.1%) and Urban (16.9%), reflecting the store's wide regional reach (see Area Type Bar Chart).



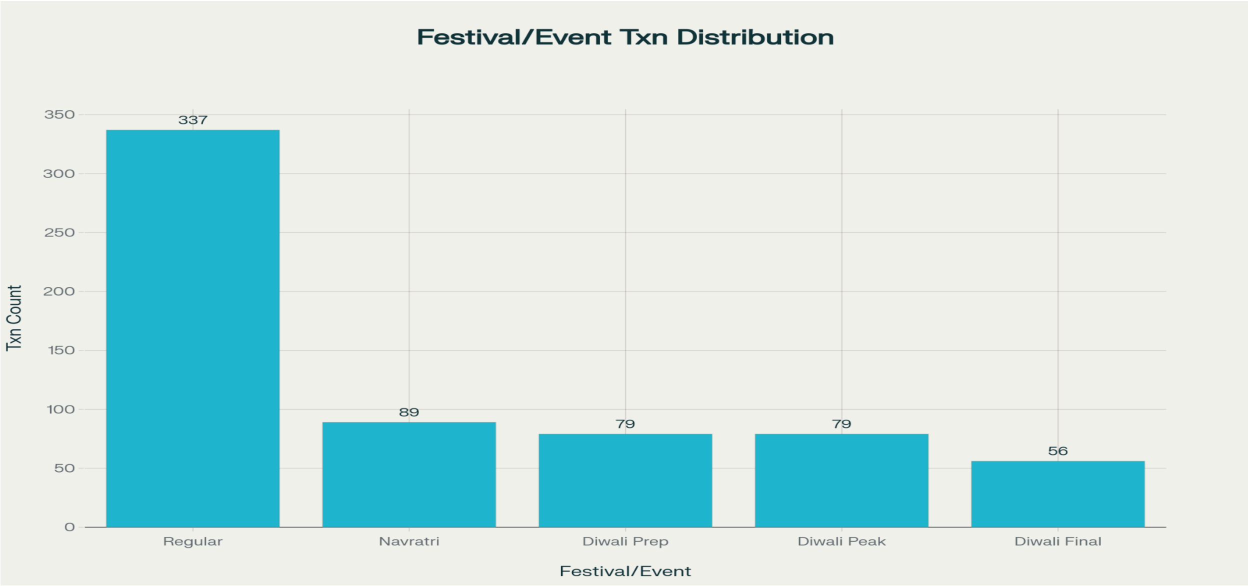
Product Category Distribution
Txn Count
200
150
100
50
0
31.1%
26.7%
20.8%
11.1%
10.3%
Niwar
Rope
Brooms
Wiper
Sutli
Prod Category

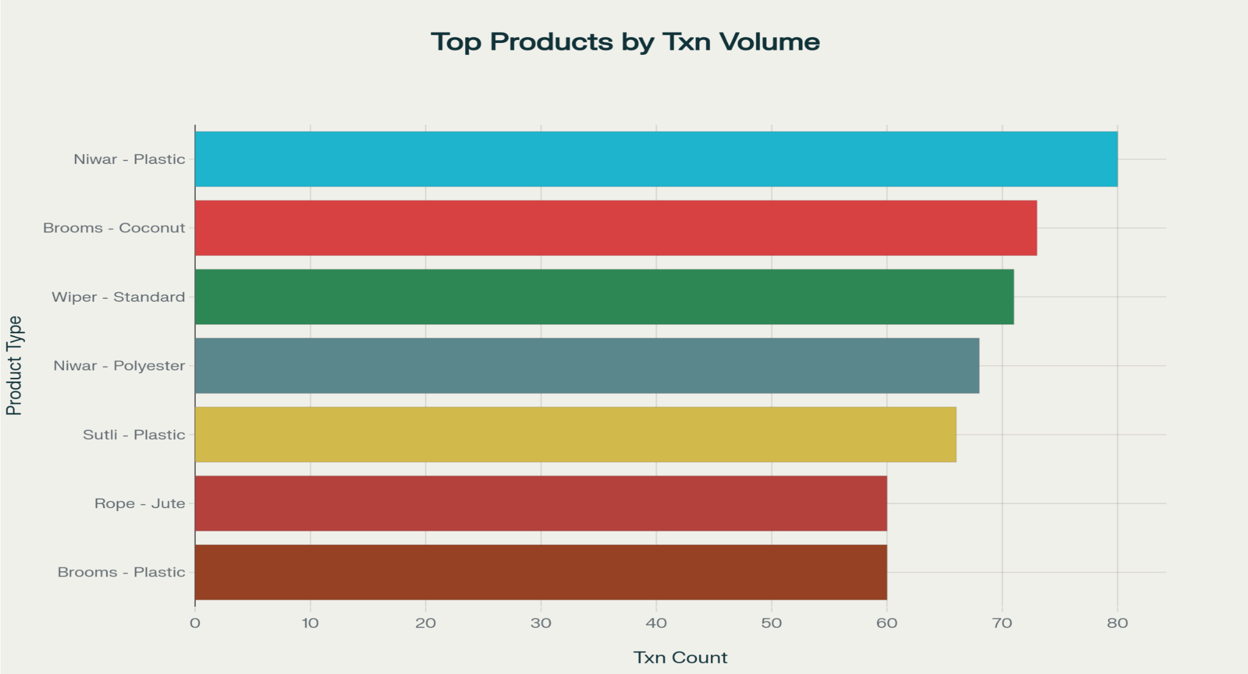


Customer Type Distribution
Wholesale
Retail
Retail
34.2%
Wholesale
65.8%



Geographical Transaction Distribution
Num of Txns
350
300
250
200
150
100
50
0
52.0%
31.1%
16.9%
Rural
Semi-Urban
Urban
Area Type





Seasonal/Festival Impact: Non-festival sales form 52.7%, while festival periods (Navratri, Diwali Prep, Peak, and Final) contribute 47.3%, confirming the strong influence of seasonality on sales volume (see Festival/Event Bar Chart)

Top-Selling Product Types: Niwar–Plastic, Broom–Coconut, and Wiper–Standard emerge as the highest-selling items, indicating fast-moving products that require stock prioritization (see Top Product Types Bar Chart).

These categorical insights visually and statistically reinforce key trends in customer demand, geographic performance, and product movement. They directly support strategic decisions on inventory allocation, marketing focus, and festival-based stock management.

4. **Detailed Explanation of Analysis Process/ Method**

The analysis of 640 transactions (July 20 – Oct 31, 2025) combined forecasting, inventory classification, customer segmentation, and profitability assessment to interpret sales patterns, optimize stock, and evaluate financial performance. The methods were selected for their accuracy, interpretability, and suitability for small-business data.

1. Seasonal ARIMA (SARIMA) Forecasting
Daily sales totals were modeled using SARIMA to capture trends and seasonality around events such as Navratri and Diwali. It was preferred over linear and exponential models for handling periodic demand and over machine-learning models like Random Forest for its lower data needs and interpretability. The forecasts guided inventory replenishment and peak-season planning.

2. ABC Inventory Classification
Products were ranked by revenue contribution, identifying Niwar and Brooms as “A” items contributing ~65% of total sales. The method prioritizes high-value products and efficient capital use, performing better than frequency- or interval-based approaches that ignore financial weight.

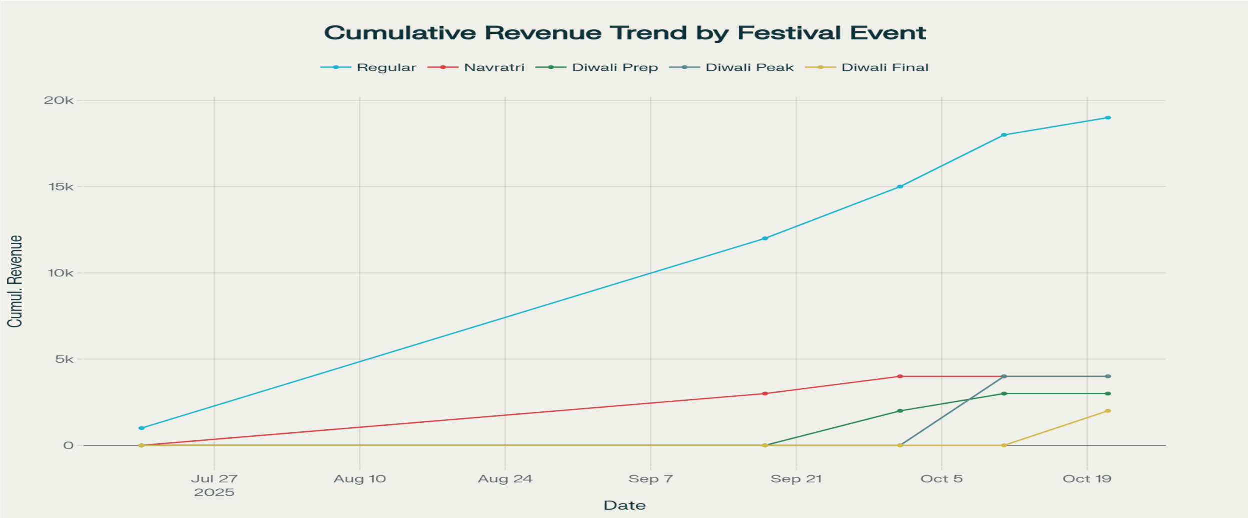
3. Customer Segmentation (K-Means Clustering)
Customers were grouped by purchase frequency, transaction value, and location, revealing clusters such as frequent wholesale and occasional retail buyers. K-Means was chosen for its speed, clear cluster boundaries, and practicality over hierarchical or density-based methods, enabling targeted marketing and stock planning.

4. Profit Margin & Spread Analysis
Margins by product, customer type, and region were analyzed to assess profitability. Wholesale margins (4–7%) and retail margins (9–12%) confirmed sustainable returns. Margin-spread analysis exposed intra-category variation and pricing opportunities, offering richer insights than gross or net ratios.

5. Results and Findings

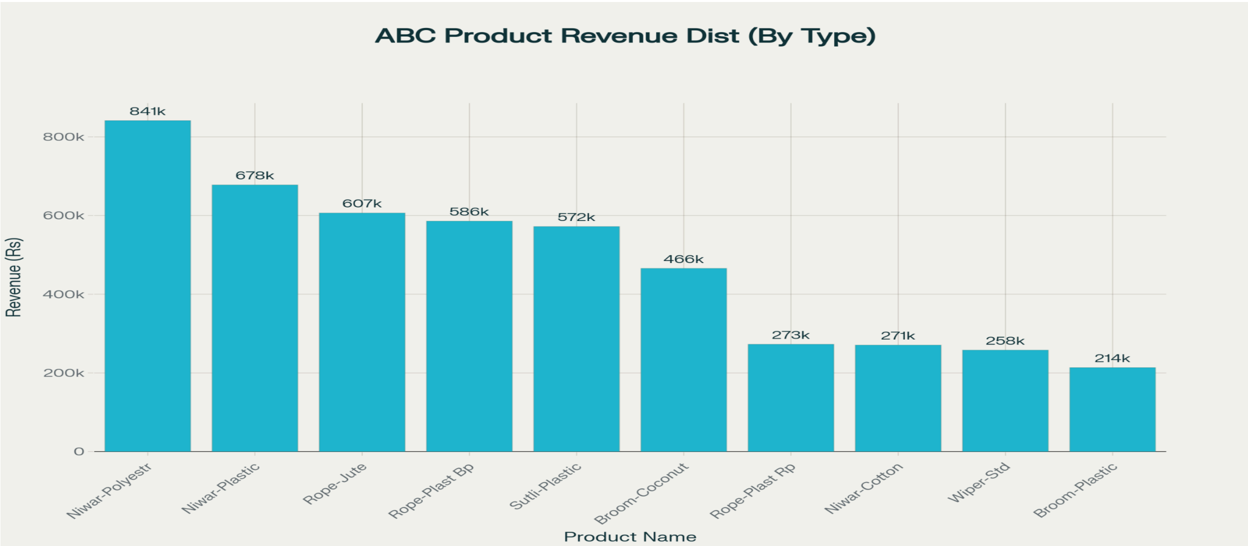
1. Strong Seasonal Demand and Festival Impact

Analysis confirms significant seasonality, with a 127% increase in revenue during the festival season (Navratri and Diwali) compared to the regular period. Festival events (Navratri, Diwali Prep, Diwali Peak, Diwali Final) together account for 47.3% of transactions, demonstrating that these occasions are critical sales drivers. Rural segments show even stronger seasonal sensitivity (+150%), underscoring the need for region-specific inventory strategies.



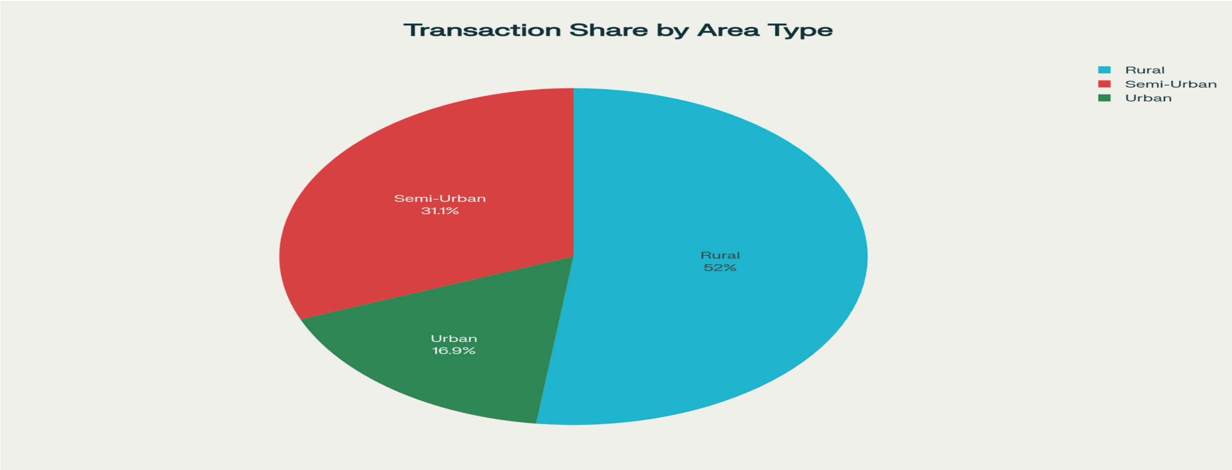
2. Product Performance and Inventory Prioritization

ABC analysis reveals that Niwar and Brooms together generate 65% of revenue, clearly marking them as fast-moving and priority inventory items. Other products like Rope, Wiper, and Sutli contribute less but maintain steady sales. This insight guides efficient allocation of procurement resources and shelf space, minimizing capital locked in slow-moving items.



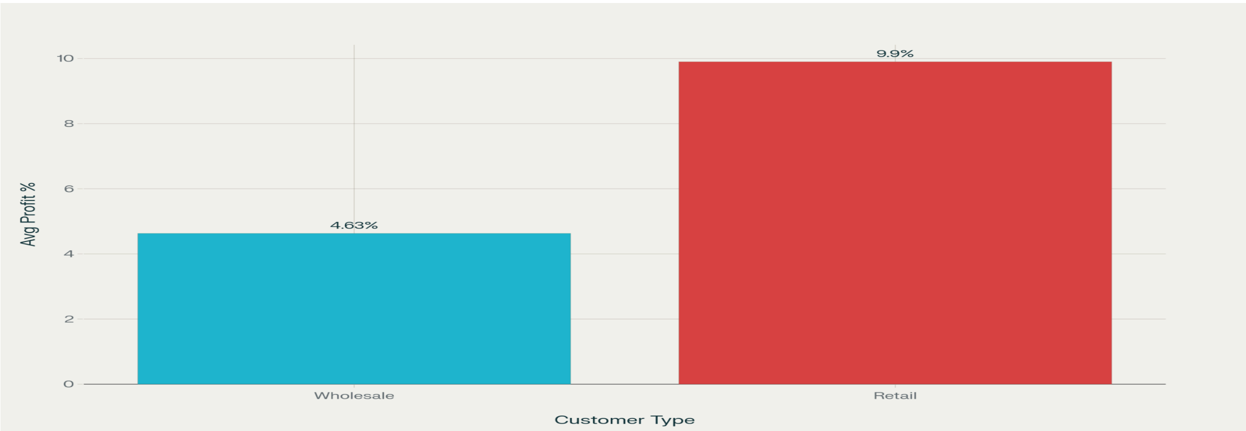
3. Customer and Market Segmentation

The dual-market model is evident in transaction composition: wholesale contributes approximately 65.8% of transactions, dominating revenue though at lower profit margins (~4-7%), whereas retail comprises 34.2%, with higher margins (9–12%). Geographic analysis shows a 52% transaction share from rural customers, highlighting the importance of rural demand in business planning.



4. Profitability Insights

Profit margin analysis affirms healthy overall profitability with average profit margins around 6.4%. The differentiation in margins between wholesale and retail customers equips the management with pricing and promotional strategies customized per segment to optimize overall margin.



5. Predictive Analysis Readiness

Preliminary seasonal trend analysis using daily sales data lays the groundwork for future application of ARIMA or other forecasting models. This can strongly support proactive inventory management and demand planning by anticipating peak sales periods.